

Complex Variables & Linear Algebra (BMAT201L)

Module 1

Analytic Functions

1 Introduction

An ordered pair of real numbers is called a *complex number*. If $z = (x, y)$ is a complex number, then x and y are real numbers and are known as the *real* and *imaginary* parts of z . The set of all complex numbers is denoted by $\mathbb{C}$. The *sum* and the *product* of two complex numbers $z_1 = (x_1, y_1)$ and $z_2 = (x_2, y_2)$ are respectively defined by

$$z_1 + z_2 = (x_1 + x_2, y_1 + y_2), \quad (1.1)$$

$$z_1 z_2 = (x_1 x_2 - y_1 y_2, x_1 y_2 + x_2 y_1). \quad (1.2)$$

We identify the set $\mathbb{R}$ of real numbers with a subset of $\mathbb{C}$ through the one-to-one correspondence $x \rightarrow (x, 0)$ and for any real x , we write $x = (x, 0)$. Set $i = (0, 1)$. Then $i^2 = (0, 1)(0, 1) = (-1, 0) = -1$ so that $i = \sqrt{-1}$, which is called the *imaginary unit*. Some times, it is also represented by the greek symbol ι . Also,

$$z = (x, y) = (x, 0) + (0, y) = (x, 0) + (0, 1)(y, 0) = x + iy, \quad (1.3)$$

Let $z = (x, y) \in \mathbb{C}$. Its conjugate is $\bar{z} = x - iy$. The *inverse* of $z = (x, y) \neq (0, 0)$ is given by

$$w = \left(\frac{x}{x^2 + y^2}, -\frac{y}{x^2 + y^2} \right). \quad (1.4)$$

Let $z = (x, y) = (r \cos \theta, r \sin \theta) \neq (0, 0)$. Then $|z|^2 = r^2 \cos^2 \theta + r^2 \sin^2 \theta = r^2$, that is

$$r = |z| = (x^2 + y^2)^{1/2}, \quad (1.5)$$

is called its *modulus* or *absolute value*, and

$$\theta = \tan^{-1} \left(\frac{y}{x} \right), \quad (1.6)$$

is called the *argument* of z and is denoted by the symbol $\arg(z)$. Since $\sin \theta$ and $\cos \theta$ are periodic functions with period 2π , we see that

$$z = r \cos \theta + i r \sin \theta = z = r \cos(\theta + 2n\pi) + i r \sin(\theta + 2n\pi)$$

for $n = 1, 2, 3, \dots$. Thus $\arg z$ is a *many-valued relation* but not a function. However, $\arg z \in (0, 2\pi]$ is called the *principal argument* $\text{Arg}(z)$ of z . We have

$$\text{Arg}(z) = \begin{cases} \tan^{-1} \frac{y}{x}, & x > 0, y > 0, \\ \pi - \tan^{-1} \frac{y}{|x|}, & x < 0, y > 0, \\ \pi + \tan^{-1} \left| \frac{y}{x} \right|, & x < 0, y < 0, \\ 2\pi - \tan^{-1} \frac{|y|}{x}, & x > 0, y < 0. \end{cases} \quad (1.7)$$

In view of Euler's identity: $e^{i\theta} = \cos \theta + i \sin \theta$, we write the *polar form* of $z = (x, y) \in \mathbb{C}$ as

$$z = x + iy = r(\cos \theta + i \sin \theta) = re^{i\theta}. \quad (1.8)$$

2 Argand Plane

Each complex number z is denoted as a point $P(x, y)$ in the XY-plane and vice versa. The X-axis representing the points of the form $(x, 0)$ is called the *real axis*, and the Y-axis representing the points of the form $(0, y)$ is called the *imaginary axis*. The distance OP of P from the origin (that is pole) $O = (0, 0)$ is the modulus $|z|$ of z and $\arg(z)$ is the angle made by the radius vector OP at O with the positive real axis OX in the positive (anticlockwise) direction. The XY-plane is called the *Argand plane* or *complex plane*.

Remark 2.1. Let $z_0 \in \mathbb{C}$ and $a > 0$, then the set

$$\gamma_a(z_0) = \{z \in \mathbb{C} : |z - z_0| = a\} \quad (2.1)$$

defines a circle with centre at z_0 and radius a .

Remark 2.2. The set $\text{ext } \gamma_a(z_0) = \{z \in \mathbb{C} : |z - z_0| > a\}$ defines the *exterior* a circle with centre at z_0 and radius a . Whereas, $\text{int } \gamma_a(z_0) = \{z \in \mathbb{C} : |z - z_0| < a\}$ defines the *interior* a circle with centre at z_0 and radius a .

3 Powers and Roots of a Complex Number

Theorem 3.1 (Powers of a Complex Number). Let $z = x + iy \in \mathbb{C}$ be represented by the polar form (1.8), and n be a positive integer. Then applying de Moivre's theorem, we have

$$z^n = r^n(\cos n\theta + i \sin n\theta) = r^n e^{in\theta}. \quad (3.1)$$

Theorem 3.2 (Fundamental Theorem of Algebra). If $P(z)$ is a polynomial of degree $n \geq 1$, then P has at least one zero.

Corollary 3.1. If $P(z)$ is a polynomial of degree $n \geq 1$ with complex coefficients, then the equation $P(z) = 0$ has precisely n roots.

Theorem 3.3 (Roots of a Complex Number). Let $c = a + ib \neq 0 \in \mathbb{C}$. The n th roots of c are given by

$$\omega_k = r^{1/n} \left[\cos \left(\frac{\theta + 2k\pi}{n} \right) + i \sin \left(\frac{\theta + 2k\pi}{n} \right) \right], k = 0, 1, \dots, n-1, \quad (3.2)$$

which are the n roots of the polynomial equation $z^n - c = 0$ or $z^n = c$.

Remark 3.1. If $k = 0$, ω_0 is known as the *principal n th root* of c .

Remark 3.2. Since the n th roots of a complex number c have the same modulus, they lie on a circle of radius $r^{1/n}$, centered at the origin $(0, 0)$ in the complex plane. Also, since the difference between the arguments of any two successive roots is $2\pi/n$, the n th roots are equally spaced on this circle.

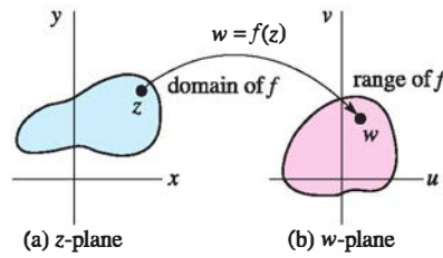
4 Complex Function of a Complex Variable

Definition 4.1. Let $A \subset \mathbb{C}$. A function $f : A \rightarrow \mathbb{C}$ is called a *complex-valued function* of complex variable z or simply a *complex function* of complex-variable z . Let $w = f(z) = u(x, y) + iv(x, y)$. Then the components $u \equiv u(x, y)$ and $v \equiv v(x, y)$ of f are the real and imaginary parts of f .

Remark 4.1. A complex valued function of a complex variable is a pair of real-valued functions of two variables.

Definition 4.2. Let $A \subset \mathbb{C}$. The graph of $f = u + iv : A \rightarrow \mathbb{C}$ of complex variable $z = x + iy$ is the set $\text{gr}(f) = \{(z, f(z)) : z \in A\}$.

Remark 4.2. Since f involves two components u and v , which are functions of two independent variables x and y , four axes in four dimensions are needed to draw the graph of f . However, a complex function f can be illustrated by a pair of planes with rectangular coordinates, namely the z -plane and the w -plane. The points $z = (x, y)$ are plotted in the z -plane, whereas the points $w = (u, v)$ are represented in the w -plane. Thus a complex function f is interpreted as a *mapping* or *transformation* from a set of points of the z -plane to the set of points of w -plane.

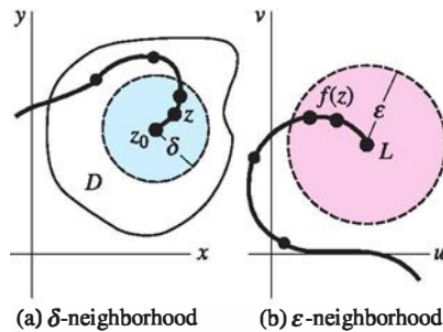


5 Limit

For $z_0 \in \mathbb{C}$ and $\delta > 0$, the $\delta > 0$ -neighborhood of z_0 is the set

$$N_\delta(z_0) = \{z \in \mathbb{C} : |z - z_0| < \delta\}. \quad (5.1)$$

A function $f : S \rightarrow \mathbb{C}$ is said to tend to a *limit* l as z approaches z_0 , if for each $\varepsilon > 0$, there exists a $\delta > 0$ such that $|f(z) - l| < \varepsilon$ whenever $0 < |z - z_0| < \delta$. We write it as $\lim_{z \rightarrow z_0} f(z) = l$.



Remark 5.1. Since z and z_0 are points in the complex plane, when $\lim_{z \rightarrow z_0} f(z) = l$ exists, we mean that $f(z)$ approaches l as the point z approaches z_0 from any direction.

Theorem 5.1 (Test for the Nonexistence of the Limit). If $f(z)$ is different along different paths of approach as $z \rightarrow z_0$, then the limit of $f(z)$ does not exist at z_0 .

6 Continuity

A function $w = f(z)$ is *continuous* at z_0 , if $\lim_{z \rightarrow z_0} f(z) = f(z_0)$, provided the limit exists. We say that f is continuous on the region S , if it is continuous at every point of S . If $f(z) = u(x, y) + iv(x, y)$, then f is continuous at $z_0 = (x_0, y_0)$ if and only if the real and imaginary parts $u(x, y)$ and $v(x, y)$ are continuous at (x_0, y_0) .

Remark 6.1. Let $f(z)$ and $g(z)$ be continuous at z_0 . Then the sum $f + g$, product fg and the composition $f \circ g$ are continuous at z_0 . The quotient f/g is continuous at z_0 , provided $g(z_0) \neq 0$. An n th degree polynomial function of the form $f(z) = a_0 z^n + a_1 z^{n-1} + \cdots + a_{n-1} z + a_n$ with complex coefficients a_i , is continuous on the entire complex plane. A rational function of the form $f(z) = p(z)/q(z)$, where p and q are polynomial functions with complex coefficients, is continuous at all points of the complex plane except at the zeros of q .

Definition 6.1 (Path and its Connectedness). Let $S \subset \mathbb{C}$. A *path* in S is a continuous function $\gamma: [a, b] \rightarrow S$, where $a, b \in \mathbb{R}$ with $a \leq b$. A *closed path* is a path that ends on its starting point ($\gamma(b) = \gamma(a)$). A set is *path-connected*, if for any $z_1, z_2 \in S$, there exists a path γ in S such that $\gamma(a) = z_1$ and $\gamma(b) = z_2$.

7 Analytic Functions

Definition 7.1 (Derivative). Let $S \subset \mathbb{C}$ and $z_0 \in S$. A function $f: S \rightarrow \mathbb{C}$ is said to be *differentiable* at z_0 , if

$$f'(z_0) = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}, \quad (7.1)$$

provided the limit (7.1) exists. If $f'(z_0)$ exists, it is called the *derivative* of f at z_0 . We say that f is differentiable on S , if it is differentiable at every point of S .

Definition 7.2 (Analytic Function). Let $S \subset \mathbb{C}$ and $z_0 \in S$. A function f is said to be *analytic* at a point $z_0 \in S$, if it is differentiable on some neighborhood of z_0 . A function $w = f(z)$ is *analytic* on S , if and only if it is analytic at every point of S .

Remark 7.1. Everywhere differentiable functions are analytic everywhere on $\mathbb{C}$. Conversely, everywhere analytic functions are everywhere differentiable.

Definition 7.3 (Entire Function). A complex function which is analytic throughout the complex plane $\mathbb{C}$ is called an *entire* function.

Remark 7.2. If $f(z)$ is not differentiable at every point of the complex plane, then it will be analytic at no point of the complex plane. That is nowhere differentiable functions are nowhere analytic.

8 Cauchy-Riemann Equations

Theorem 8.1 (Necessary Conditions for Differentiability). Let $D \subset \mathbb{C}$ and $z_0 = (x_0, y_0)$. If $f = u + iv: D \rightarrow \mathbb{C}$ is differentiable at z_0 , that is if $f'(z_0)$ exists, then

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \text{ and } \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \text{ at } (x_0, y_0). \quad (8.1)$$

The conditions (8.1) are called the *Cauchy-Riemann* equations.

Remark 8.1. If the Cauchy-Riemann equations are not satisfied at a point, f cannot be differentiable at that point. Thus the conditions (8.1) are *necessary* for a function $f(z)$ to be differentiable at a point z_0 .

Example 8.1. Show that $f(z) = \operatorname{Re}(z) = x$ for all $z \in \mathbb{C}$ is nowhere differentiable and hence, nowhere analytic.

Solution. Write $u \equiv u(x, y) = x$ and $v \equiv v(x, y) = 0$. Then $\frac{\partial u}{\partial x} = 1$, $\frac{\partial u}{\partial y} = 0$, $\frac{\partial v}{\partial x} = 0$ and $\frac{\partial v}{\partial y} = 0$ at every $z = (x, y) \in \mathbb{C}$. Thus $\frac{\partial u}{\partial x} \neq \frac{\partial v}{\partial y}$. That is the Cauchy-Riemann equations (8.1) are not satisfied at every $z \in \mathbb{C}$. Therefore, f is nowhere differentiable and hence nowhere analytic.

Example 8.2. The following functions are nowhere differentiable and hence nowhere analytic:

- (a) $f(z) = \operatorname{Im}(z) = y$,
- (b) $f(z) = \bar{z} = x - iy$,
- (c) $f(z) = x(3 + 2iy^2)$.

Solution.

- (a) Write $u \equiv u(x, y) = 0$ and $v \equiv v(x, y) = y$. Then $\frac{\partial u}{\partial x} = 0$, $\frac{\partial u}{\partial y} = 0$, $\frac{\partial v}{\partial x} = 0$ and $\frac{\partial v}{\partial y} = 1$ at every $z = (x, y) \in \mathbb{C}$. Thus $\frac{\partial u}{\partial x} \neq \frac{\partial v}{\partial y}$. That is the Cauchy-Riemann equations (8.1) are not satisfied at every $z \in \mathbb{C}$. Therefore, f is nowhere differentiable and hence nowhere analytic.
- (b) Write $u \equiv u(x, y) = x$ and $v \equiv v(x, y) = -y$. Then $\frac{\partial u}{\partial x} = 1$, $\frac{\partial u}{\partial y} = 0$, $\frac{\partial v}{\partial x} = 0$ and $\frac{\partial v}{\partial y} = -1$ at every $z = (x, y) \in \mathbb{C}$. Thus $\frac{\partial u}{\partial x} \neq \frac{\partial v}{\partial y}$. That is the Cauchy-Riemann equations (8.1) are not satisfied at every $z \in \mathbb{C}$. Therefore, f is nowhere differentiable and hence nowhere analytic.
- (c) Write $u \equiv u(x, y) = 3x$ and $v \equiv v(x, y) = 2xy^2$. Then $\frac{\partial u}{\partial x} = 3$, $\frac{\partial u}{\partial y} = 0$, $\frac{\partial v}{\partial x} = 2y^2$ and $\frac{\partial v}{\partial y} = 4xy$ at every $z = (x, y) \in \mathbb{C}$. Now, the Cauchy-Reimann equations (8.1) hold good only if

$$3 = 4xy \text{ and } 0 = 2y^2. \quad (8.2)$$

Note that the second relation of (8.2) gives $y = 0$. Substituting this in the first relation of (8.2), we get $3 = 0$, which is not possible. Thus Cauchy-Reimann equations (8.1) are not satisfied at every $z = (x, y) \in \mathbb{C}$. Therefore, f is nowhere differentiable and hence nowhere analytic.

Exercise 8.1. Show that the following functions are nowhere differentiable and hence nowhere analytic:

- (a) $f(z) = y + ix$,
- (b) $f(z) = -x + iy + 5$.

Remark 8.2. Cauchy-Riemann equations (8.1) are not sufficient for a function f to be differentiable at a point z_0 . That is, a complex function f may satisfy (8.1) at a point, without being differentiable at that point.

Example 8.3 (Non-differentiable Function, satisfying Cauchy-Riemann equations (8.1) at a Point). Show that

$$f(z) = \begin{cases} \left(\frac{x^3 - 3xy^2}{x^2 + y^2} \right) + i \left(\frac{y^3 - 3x^2y}{x^2 + y^2} \right), & \text{if } z \neq 0 \\ 0, & \text{if } z = 0. \end{cases} \quad (8.3)$$

is not differentiable at $z = 0$, though Cauchy-Riemann equations (8.1) hold good at the origin.

Solution. From (8.3), we see that

$$L = \frac{f(z) - f(0)}{z - 0} = \frac{x^3 - 3xy^2}{(x^2 + y^2)(x + iy)} + i \frac{y^3 - 3x^2y}{(x^2 + y^2)(x + iy)}. \quad (8.4)$$

Along the real axis, $y = 0$ and $x \rightarrow 0$ as $z \rightarrow 0$. With these, (8.4) becomes

$$L = \frac{x^3}{x^2 \cdot x} + i \frac{0}{x^2 \cdot x} = 1. \quad (8.5)$$

Whereas, along the straight line $y = x$, $x \rightarrow 0$ as $z \rightarrow 0$. With these, (8.4) becomes

$$L = \frac{x^3 - 3x \cdot x^2}{(x^2 + x^2)(x + ix)} + i \frac{x^3 - 3x^2 \cdot x}{(x^2 + x^2)(x + ix)} = -\frac{1}{1+i} - i \frac{1}{1+i} = -4. \quad (8.6)$$

From (8.5) and (8.6), we find that L approaches different numbers along different of approach as $z \rightarrow 0$. Hence, f is not differentiable at 0.

We now verify whether the Cauchy-Riemann equations (8.1) hold good at the origin: Recall that the first order partial derivatives of a function $\omega(x, y)$ of two variables at a point (a, b) are given by

$$\frac{\partial \omega}{\partial x} = \lim_{h \rightarrow 0} \frac{\omega(a+h, b) - \omega(a, b)}{h}, \quad \frac{\partial \omega}{\partial y} = \lim_{k \rightarrow 0} \frac{\omega(a, b+k) - \omega(a, b)}{k}.$$

From (8.3), write $u(x, y) = \frac{x^3 - 3xy^2}{x^2 + y^2}$ and $v(x, y) = \frac{y^3 - 3x^2y}{x^2 + y^2}$. Then

$$\begin{aligned} \frac{\partial u}{\partial x} &= \lim_{h \rightarrow 0} \frac{u(h, 0) - u(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{1}{h} \left(\frac{h^3 - 0}{h^2 + 0} \right) = 1, \\ \frac{\partial u}{\partial y} &= \lim_{k \rightarrow 0} \frac{u(0, k) - u(0, 0)}{k} = \lim_{k \rightarrow 0} \frac{1}{k} \left(\frac{0 - 0}{0 + k^2} \right) = 0, \\ \frac{\partial v}{\partial x} &= \lim_{h \rightarrow 0} \frac{v(h, 0) - v(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{1}{h} \left(\frac{0 - 0}{h^2 + 0} \right) = 0, \\ \frac{\partial v}{\partial y} &= \lim_{k \rightarrow 0} \frac{v(0, k) - v(0, 0)}{k} = \lim_{k \rightarrow 0} \frac{1}{k} \left(\frac{k^3 - 0}{0 + k^2} \right) = 1. \end{aligned}$$

That is the Cauchy-Riemann equations (8.1) hold good at the origin.

Exercise 8.2. Show that each of the following functions $f(z)$ satisfies the Cauchy Riemann equations (8.1) at the origin, but the derivative $f'(0)$ of f at $z = 0$ does not exist:

$$(a) f(z) = \begin{cases} \frac{x^2 y^5 (x+iy)}{x^4 + y^{10}}, & z \neq 0 \\ 0, & z = 0, \end{cases}$$

$$(b) f(z) = \begin{cases} \frac{x^3 - y^3}{x^2 + y^2} + i \left(\frac{x^3 + y^3}{x^2 + y^2} \right), & z \neq 0 \\ 0, & z = 0, \end{cases}$$

$$(c) f(z) = \begin{cases} \frac{x^3 y^2}{(x^2 + y^2)^2}, & z \neq 0 \\ 0, & z = 0. \end{cases}$$

Theorem 8.2 (Sufficient Conditions for Differentiability). Consider $f(z) = u(z) + iv(z)$ and $z_0 = (x_0, y_0) \in \mathbb{C}$. Suppose that

(a) the partial derivatives $\frac{\partial u}{\partial x}$, $\frac{\partial u}{\partial y}$, $\frac{\partial v}{\partial x}$ and $\frac{\partial v}{\partial y}$ are continuous at (x_0, y_0) , and

(b) satisfy the Cauchy-Riemann equations: $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$.

Then f will be differentiable at z_0 , and its derivative is given by

$$f'(z_0) = \frac{\partial u}{\partial x}(x_0, y_0) + i \frac{\partial v}{\partial x}(x_0, y_0) = \frac{\partial v}{\partial y} - i \frac{\partial u}{\partial y}. \quad (8.7)$$

Remark 8.3 (Analytic Functions are Differentiable). If $f(z)$ is analytic at a point z_0 , then it is necessarily differentiable at a point. However, the converse is not true. In fact, there are functions which are differentiable at a point but are nowhere analytic.

Example 8.4 (Nowhere Analytic Function, which is Differentiable at only one Point). Show that $f(z) = |z|^2 = x^2 + y^2$ is differentiable at $z = 0$, but not analytic there at. In fact, f is nowhere analytic.

Solution. The real and imaginary parts of f are $u(x, y) = x^2 + y^2$ and $v(x, y) = 0$. Then the first order partial derivatives of u and v with respect to x and y are given by

$$\frac{\partial u}{\partial x} = 2x, \frac{\partial u}{\partial y} = 2y, \frac{\partial v}{\partial x} = 0, \frac{\partial v}{\partial y} = 0.$$

Cauchy Riemann equations (8.1) hold good only if $x = 0 = y$. Moreover, $\frac{\partial u}{\partial x}$, $\frac{\partial u}{\partial y}$, $\frac{\partial v}{\partial x}$ and $\frac{\partial v}{\partial y}$ are continuous at $z = 0$. Therefore, by Theorem 8.2, f is differentiable at $z = 0$, and the derivative of f at $z = 0$ is

$$f'(0) = \frac{\partial u}{\partial x}(0, 0) + i \frac{\partial v}{\partial x}(0, 0) = 0 + i0 = 0.$$

It is obvious that the Cauchy-Riemann equations (8.1) are not satisfied at $z \neq (0, 0)$, and hence f is not differentiable at $z \neq (0, 0)$. Since there exists no neighborhood of any $z \in \mathbb{C}$, in which f is differentiable, f is nowhere analytic.

Exercise 8.3. Show that each of the following functions $f(z)$ is differentiable only at the origin but nowhere analytic. Also, find $f'(0)$ in each case.

(a) $x^3 - xy^2 - 2x^2y^2 + i(2x^2y - y^3),$

(b) $x^2 - ixy.$

Example 8.5 (Nowhere Analytic Function, which is Differentiable along a Curve). Show that $f(z) = x^2y^2$ is differentiable along the coordinate axes, but is nowhere analytic.

Solution. Write $u(x, y) = x^2y^2$, $v(x, y) = 0$. Then $\frac{\partial u}{\partial x} = 2xy^2$, $\frac{\partial u}{\partial y} = 2x^2y$, $\frac{\partial v}{\partial x} = 0$, $\frac{\partial v}{\partial y} = 0$ are continuous on $\mathbb{C}$ and hence at $z = 0$. Since, Cauchy Riemann equations (8.1) hold good only if $x = 0$ or $y = 0$, by Theorem 8.2, f is differentiable on the coordinate axes. The derivative of f along the coordinate axes is

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = 2xy^2 + i0 = 0.$$

Since there is no neighborhood of any of the points in the complex plane, on which f is differentiable. Hence f is nowhere analytic.

Exercise 8.4. Show that each of the following functions is nowhere analytic, but is differentiable along the the coordinate axes. Also, find $f'(z)$ in each case:

(a) $f(z) = 3x^2y^2 - i6x^2y^2,$

(b) $f(z) = x^3 + 3xy^2 - x + i(y^3 + 3x^2 - y).$

9 Entire Functions

Recall that a complex function which is analytic throughout the complex plane $\mathbb{C}$ is an *entire* function.

Example 9.1. Show that each of the following is an entire function, and find the derivative at each $z \in \mathbb{C}$:

(a) $f(z) = e^x \cos y + i e^x \sin y,$

(b) $f(z) = x^3 - 3xy^2 + i(3x^2y - y^3),$

(c) $f(z) = \sin x \cosh y + i \cos x \sinh y.$

Solution.

(a) Let $u(x, y) = e^x \cos y$ and $v(x, y) = e^x \sin y$. Then $\frac{\partial u}{\partial x} = e^x \cos y$, $\frac{\partial u}{\partial y} = -e^x \sin y$, $\frac{\partial v}{\partial x} = e^x \sin y$, $\frac{\partial v}{\partial y} = e^x \cos y$. Thus at every point $z = (x, y)$ the Cauchy-Reimann equations (8.1) hold good. Also, these first order partial derivatives are continuous on the entire complex plane. Thus, by Theorem 8.2, $f = u + iv$ is analytic throughout the complex plane, and

$$\begin{aligned} f'(z) &= \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = e^x \cos y + i e^x \sin y \\ &= e^x (\cos y + i \sin y) = e^{x+iy} = e^z \text{ for all } z \in \mathbb{C}. \end{aligned}$$

(b) Write $u(x, y) = x^3 - 3xy^2$ and $v(x, y) = 3x^2y - y^3$. Then

$$\frac{\partial u}{\partial x} = 3x^2 - 3y^2, \quad \frac{\partial u}{\partial y} = -6xy,$$

$$\frac{\partial v}{\partial x} = 6xy, \quad \frac{\partial v}{\partial y} = 3x^2 - 3y^2.$$

Thus at every point $z = (x, y)$ (8.1) hold good. Also, these first order partial derivatives are continuous on the entire complex plane. Thus, by Theorem 8.2, $f = u + iv$ is entire, and

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = (3x^2 - 3y^2) + i(6xy) = 3z^2.$$

(c) We write $u(x, y) = \sin x \cosh y$ and $v(x, y) = \cos x \sinh y$. Then

$$\frac{\partial u}{\partial x} = \cos x \cosh y, \quad \frac{\partial u}{\partial y} = \sin x \sinh y,$$

$$\frac{\partial v}{\partial x} = -\sin x \sinh y, \quad \frac{\partial v}{\partial y} = \cos x \cosh y.$$

Thus at every point $z = (x, y)$ the Cauchy-Reimann equations (8.1) hold good. Also, these first order partial derivatives are known to be continuous in the entire complex plane. Thus, by Theorem 8.2, $f = u + iv$ is entire, and

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = \cos x \cosh y - i \sin x \sinh y.$$

Exercise 9.1. Explain why $f(z) = x + \sin x \cosh y + i(y + \cos x \sinh y)$ is analytic at every point of the complex plane.

Example 9.2. Find the choice of the constants a and b such that

$$f(z) = (3x + ay + 5) + i(2x + by - 3)$$

is entire, and find $f(z)$.

Solution. Write $u(x, y) = 3x + ay + 5$ and $v(x, y) = 2x + by - 3$. Then

$$\frac{\partial u}{\partial x} = 3, \quad \frac{\partial u}{\partial y} = a; \quad \frac{\partial v}{\partial x} = 2, \quad \frac{\partial v}{\partial y} = b.$$

Using these values in (8.1), we see that $a = -2, b = 3$.

Thus the entire function is $f(z) = (3x - 2y + 5) + i(2x + 3y - 3)$, and

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = 3 + i2 = 3 + 2i.$$

Exercise 9.2. Find the the constants a and b such that $f(z) = e^x [\cos ay + i \sin(y + b)]$ is entire, and find $f(z)$.

Example 9.3. Show that $f(z) = \frac{x^3+xy^2+x}{x^2+y^2} + i\frac{x^2y+y^3-y}{x^2+y^2}$ is analytic in an appropriate domain, and find its derivative in the domain:

Solution. Write

$$u(x, y) = \frac{x^3+xy^2+x}{x^2+y^2} = x + \frac{x}{x^2+y^2}, \quad v(x, y) = \frac{x^2y+y^3-y}{x^2+y^2} = y - \frac{y}{x^2+y^2}.$$

Then u and v are rational functions and are continuous, provided $x^2+y^2 \neq 0$. That is u and v are continuous on $\mathbb{C} - \{0\}$. If $P = P(x, y)$ and $Q = Q(x, y)$, we recall the quotient rule of partial differentiation:

$$\frac{\partial}{\partial x} \left(\frac{P}{Q} \right) = \frac{Q \frac{\partial P}{\partial x} - P \frac{\partial Q}{\partial x}}{Q^2}, \quad \frac{\partial}{\partial y} \left(\frac{P}{Q} \right) = \frac{Q \frac{\partial P}{\partial y} - P \frac{\partial Q}{\partial y}}{Q^2}. \quad (9.1)$$

to find the first order partial derivatives at $z \neq 0$:

$$\begin{aligned} \frac{\partial u}{\partial x} &= 1 + \frac{\partial}{\partial x} \left\{ \frac{x}{x^2+y^2} \right\} = \frac{(x^2+y^2)(1) - (2x)(x)}{(x^2+y^2)^2} = 1 + \frac{y^2-x^2}{(x^2+y^2)^2}, \\ \frac{\partial u}{\partial y} &= x \frac{\partial}{\partial x} \left\{ \frac{1}{x^2+y^2} \right\} = -\frac{2xy}{(x^2+y^2)^2}, \\ \frac{\partial v}{\partial x} &= -y \frac{\partial}{\partial x} \left\{ \frac{1}{x^2+y^2} \right\} = \frac{2xy}{(x^2+y^2)^2}, \\ \frac{\partial v}{\partial y} &= 1 - \frac{\partial}{\partial y} \left\{ \frac{y}{x^2+y^2} \right\} = 1 - \frac{(x^2+y^2)(1) - (2y)(y)}{(x^2+y^2)^2} = 1 + \frac{y^2-x^2}{(x^2+y^2)^2}. \end{aligned}$$

Thus (8.1) are satisfied at $z = (x, y) \neq (0, 0)$. In view of Theorem 8.2 $f = u + iv$ is analytic on $\mathbb{C} - \{0\}$, and

$$\begin{aligned} f'(z) &= \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = \left\{ 1 + \frac{y^2-x^2}{(x^2+y^2)^2} \right\} + i \left\{ 1 + \frac{2xy}{(x^2+y^2)^2} \right\} \\ &= (1+i) - \frac{x^2-y^2-2ixy}{(x^2+y^2)^2} = (1+i) - \frac{(x-iy)^2}{(x-iy)^2((x+iy)^2)} = (1+i) - \frac{1}{z^2}, \quad z \neq 0. \end{aligned}$$

Exercise 9.3. Prove or disprove the following statements:

- Everywhere differentiable functions are analytic everywhere,
- If $f = u + iv$ satisfies the Cauchy-Riemann equations (8.1) at every point of the complex plane, then it will be differentiable at at least one point of the complex plane,
- Nowhere analytic function is nowhere differentiable,
- If f is differentiable at a point of the complex plane, then it will be analytic at that point.